

Theory Plan — Phase-Reconfigurable ADMM Accelerator

Everything here is constrained by what the silicon and the bit-exact model already told us. No result is planned that the data contradicts.

0. What the measurements force us to accept

These are not negotiable. Every proof below must be consistent with them.

#	Measurement	Consequence for the theory
M1	Fabric: prox array 2720 $\rightarrow$ 1712 LUTs (−37.1%). With DSP: 1392 $\rightarrow$ 1432 (+2.9%)	Area saving is conditional on the arithmetic being in fabric. Any theorem predicting unconditional saving is false.
M2	Fabric multiplier: 166 LUTs @18×18, 35 @9×9, ratio 4.74 (W ² predicts 4.00)	Lane area is dominated by a quadratic multiplier term.
M3	Casts cost ~40 LUTs/8 lanes with no offsetting saving under DSP	Narrowing has a fixed overhead that must appear in the cost model.
M4	Required lane widths are 6–9 fractional bits, not 16	The Q2.16 baseline is over-provisioned; the "uniform 18-bit baseline" is a weak strawman unless justified.
M5	Ordering $W^*_L1 > W^*_{Box} \geq W^*_L2$ fails under NMSE at every threshold; holds under abs-error in some regimes; Box is consistently <i>cheapest</i>	The ordering is not a fixed inequality. It is a function of problem statistics.
M6	L1 active-set misclassification = 0 for $F \geq 4$	The old Theorem 2 mechanism is not operative. Its premise must go.
M7	Binding overflow is $\boxed{w = q + \rho(z-u)}$, peak 3.80 unscaled	Range constraint is on the pre-combination, not on the L2 lane.
M8	$\max M \leq 1/\rho$; measured 2.116 at $\rho=0.125$, 1.117 at $\rho=0.5$	A lower bound on ρ exists, derived from M's spectrum — not from γ .

1. The reframe

Old claim. Proximal operators have intrinsically different precision needs; exploit this to prune; get 52.3%.

Dead. M5 kills the fixed ordering, M6 kills its mechanism, M1 kills the unconditional area claim, and the 52.3% never came from synthesisable RTL.

New claim. Two analytic models — one for how much precision each proximal operator needs, one for what a bit costs in a given fabric — compose into a closed-form predictor of whether asymmetric wordlength allocation is profitable at all, and by how much. The predictor is validated against routed silicon, and it correctly predicts both the 37.1% win in fabric and the 2.9% loss on hard macros.

This is stronger than the old claim because it says **when not to do this**, which is the part practitioners actually need and which the WLO literature — being search-based — cannot express.

2. Proof inventory

Dependency order. Each has a stated *purpose*, *method*, and *falsifier*.

Lemma 1 — Homogeneity of the fixed-point recursion

Statement. For $P \in \{\Pi_{[lo,hi]}, S_{\kappa}, \gamma\}$, scaling (q, κ, lo, hi) by $\alpha > 0$ scales (x, z, u) by α and leaves the iterate trajectory, iteration count and convergence behaviour identical. M is unchanged.

Method. Direct: show each prox is positively homogeneous of degree 1 under simultaneous scaling of its parameters, then induct over the three ADMM updates.

Purpose. Licenses the α -normalisation that made every experiment valid (M7). Without it, the scaling looks like a fudge; with it, it's exact.

Falsifier. Any operator whose parameters don't scale (e.g. a fixed saturation unrelated to the signal). State the class it covers and stop there.

Lemma 2 — Neighborhood convergence under bounded prox error

Statement. If the z -update is computed as $\hat{z} = P(v) + e_k$ with $\sup_k \|e_k\|_2 \leq \bar{e}$, then ADMM iterates converge to a ball around the exact solution of radius $C \cdot \bar{e}$, with C depending on ρ and the spectrum of M .

Method. Do **not** reprove this. Cite and adapt: Eckstein & Bertsekas (1992) for summable-error convergence, and the perturbed/inexact ADMM neighborhood results. Our contribution starts at Theorem 1, not here.

Purpose. Converts "quantization error" into "distance from optimum", which is the only way a wordlength bound means anything.

Falsifier. None expected; this is textbook. Risk is misciting — check the exact hypotheses (convexity, closedness, $\rho > 0$) hold for all three operators.

Theorem 1 — Operator error decomposition (THE CORE RESULT)

Statement. Let P_θ be a proximal operator with parameter θ , evaluated at fractional wordlength W (step $q = 2^{-W}$). Partition the input index set into the **degenerate set** D (where $\partial P/\partial v = 0$ — the output is independent of the input) and its complement D^c . Then

$$E\|e_P\|^2 = (|D^c|/n) \cdot L^2 \cdot q^2/3 + \|\partial P/\partial \theta\|^2 \cdot q_\theta^2/3 + O(q^3)$$

where L is the local Lipschitz gain on D^c and q_θ the parameter step.

Method.

Model input quantization as $e_v \sim U(-q/2, q/2)$, i.i.d., zero mean, variance $q^2/12$ per element (fix the factor-of-2 error in the old Theorem 1).
On D , $P(v + e_v) = P(v)$ exactly — the perturbation is annihilated. *This is the whole point and it is elementary to prove per operator.*
On D^c , first-order expansion gives $L \cdot e_v$.
Parameter quantization contributes independently.
Sum variances (independence), take expectation over the index partition.

Purpose. Replaces both the vacuous old Theorem 2 (active-set floor) and the hand-waved old Theorem 3. It is the single mechanism that explains all three operators *and* their regime dependence.

Why it's defensible. It's a variance calculation, not an inequality chain. Each term is separately measurable in the bit-exact model.

Falsifier. Measure $E\|e_P\|^2$ directly in `fxp_admm.py` per operator per W and check it tracks the formula. If the $(|D^c|/n)$ scaling doesn't appear, the theorem is wrong. **Run this before writing a word of it.**

Corollary 1a — Per-operator instantiation

Apply Theorem 1 to each lane:

operator	D (degenerate set)	L on D^c	θ term
Box $\Pi_{[lo,hi]}$ clipped components, $ v \geq hi$		none (bounds are exact powers of 2 in Q-format)	
L1 S_κ	zeroed components, $ v \leq \kappa - 1$	$\partial/\partial \kappa = 1$ on $D^c \rightarrow$ doubles the D^c variance	
L2 $\gamma \cdot$	$\emptyset$	γ	$\partial/\partial \gamma = v \rightarrow E[v^2] \cdot q_\gamma^2/3$, not attenuated

Immediate predictions:

Box is cheapest **when its clipped fraction is large** (BPSK detection near the bounds $\rightarrow$ most components clip $\rightarrow$ most error annihilated). Matches M5.

L1 costs $\approx \frac{1}{2}$ bit more than Box at equal degenerate fraction, plus more whenever κ is small (few components zeroed $\rightarrow |D|$ small). Matches M5.

L2 has a **floor** from coefficient quantization that contraction does not attenuate. This is exactly what the old theory missed and why L2 never came out cheapest. Matches M5.

Theorem 2 — The ordering is a function of problem statistics

Statement. Define the degenerate fractions $d_{\text{Box}} = |D_{\text{Box}}|/n$ and $d_{\text{L1}} = |D_{\text{L1}}|/n$. Then

$W^*_{\text{Box}} < W^*_{\text{L1}} \Leftrightarrow d_{\text{Box}} > d_{\text{L1}} + (\frac{1}{2} \text{ bit correction})$

and W^*_{L2} is bounded below independently of γ by the coefficient term. No fixed ordering among the three holds for all $(\kappa, \gamma, \text{box width, channel conditioning})$.

Method. Set each operator's $E\|e_P\|^2$ equal to the tolerance implied by Lemma 2 and solve for W .

Purpose. This is the honest replacement for the old Theorem 3. It *derives* the regime dependence we measured instead of asserting an inequality that fails.

Presentational note. Do not frame this as "our earlier ordering was wrong." Frame it as: the ordering is a *consequence* of operator geometry and problem statistics, and we give the condition. That is a stronger paper and it is true.

Falsifier. The $\kappa / \gamma / \text{box-width} / \text{conditioning}$ sweep must trace the predicted crossover.

`sweep_ordering.py` already runs it; add d_{Box} and d_{L1} as measured outputs and check the crossover lands where predicted.

Proposition 1 — Range feasibility and the penalty lower bound

Statement. With l integer bits (range $\pm 2^{(l-1)}$):

(a) *Weight feasibility.* $\max |M_{ij}| \leq \|M\|_2 = 1/(\lambda_{\min}(H^T H) + \rho) \leq 1/\rho$. Requiring M to be representable gives $\rho \geq \rho_{\min} = 2^{-(l-1)}$.

(b) *Pre-combination feasibility.* $\|w\|_\infty \leq \|q\|_\infty + \rho(\|z\|_\infty + \|u\|_\infty) < 2^{(l-1)}$, with $\|z\|_\infty$ bounded by the operator (exactly `hi` for Box) and $\|u\|_\infty$ bounded by dual boundedness.

Method. (a) is a one-line spectral bound. (b) is the triangle inequality plus per-operator range bounds, closed by Lemma 1's α .

Purpose. Replaces the broken old Proposition 1 wholesale. Note it *recovers the original intuition* — a lower bound on ρ — by a completely different and correct route. The old derivation had $\gamma = \rho/(\rho + \lambda)$ increasing in ρ and then called the resulting upper bound a lower bound.

Falsifier. Already validated: $l = 2$ predicts $p \geq 0.5$; measured $\max|M| = 2.116$ at $p = 0.125$ (infeasible) and 1.117 at $p = 0.5$ (feasible). **This prediction was confirmed before the theorem was written.** Say so.

Theorem 3 — Fabric cost model

Statement. Lane area in LUTs at operand width W :

$$A(W) = a \cdot W^2 \cdot 1[\text{multiplier in fabric}] + b \cdot W + c_{\text{cast}} + c_0$$

with $a \approx 0.5$ LUT/bit² measured, and the indicator collapsing to 0 whenever $W \leq W_{\text{macro}}$ and the tool infers a hard macro.

Method. Fit against the four-build sweep (M2), then validate by predicting a held-out width. **Do not fit and report on the same points.** Synthesise a third width (e.g. 13 bits) and check the model predicts it.

Purpose. Turns “narrower is smaller” from folklore into a number.

Falsifier. The held-out width. If $A(13)$ is mispredicted by more than $\sim 10\%$, the quadratic form is wrong — likely because carry-chain and cast terms don't separate cleanly.

Theorem 4 — Profitability condition (*THE HEADLINE*)

Statement. Asymmetric allocation of lane widths $\{W_{\text{op}}\}$ against a uniform baseline W_{max} is profitable iff

$$\sum_{\text{op}} [a(W_{\text{max}}^2 - W_{\text{op}}^2) \cdot 1[\text{fabric}] + b(W_{\text{max}} - W_{\text{op}})] > \sum_{\text{op}} c_{\text{cast}}$$

On a device where every $W_{\text{op}} \leq W_{\text{macro}}$, the indicator vanishes for all lanes, the quadratic term disappears, and the condition **cannot** be met — asymmetric allocation is provably net-negative.

Method. Substitute Theorem 3 into the difference of two designs. Two lines.

Purpose. This is the paper's contribution in one inequality. It predicts $+37.1\%$ and -2.9% from the same formula with the indicator flipped.

Falsifier. The four-build table. Already consistent. Strengthen by predicting a *fifth* configuration before synthesising it.

Theorem 5 — Composition

Feed W^*_{op} from Theorem 2 into Theorem 4 to get predicted area saving from problem statistics alone, with no synthesis in the loop. Compare against the measured 37.1% .

Purpose. This is what makes it *co-design* rather than two separate observations. It's also the strongest thing in the paper if it lands within a few percent.

Risk. This is the one most likely to fail. If prediction and measurement disagree badly, demote it to a discussion section and let Theorems 2 and 4 stand independently. **Do not force it.**

3. Derivation procedure — order of operations

Strict order. Do not start writing LaTeX until step 4.

Instrument Theorem 1 numerically first. Add per-operator $E||e_P||^2$ measurement to `fxp_admm.py`, split into the D and D^c contributions, and the θ term. Confirm the three-term decomposition holds before proving it. *If the decomposition doesn't appear in the data, stop and rethink.*

Measure d_Box, d_L1 across the sweep, confirm the Theorem 2 crossover.

Re-derive the lane widths from Theorem 2 + the measured tolerance. Replace 18/10/9 with the justified triple. Re-run synthesis on those.

Fit Theorem 3, then hold out a width and validate.

Write the proofs. In order: L1, L2 (cite), T1, C1a, T2, P1, T3, T4, T5.

Re-run everything end to end on the final widths and regenerate every number in the paper from one scripted pass.

Rule for the whole project: **no number enters the manuscript that isn't produced by a script in this repo.** That is how the 52.3% happened.

4. Paper flow

§	Content	Source
I	Intro: WLO is search-based, slow, gives no <i>when-not-to</i> guidance	lit review
II	Phase-reconfigurable MIMO pipeline; why one IP time-multiplexes three operators; why an N×N array serves N users not N antennas	architecture
III	Error model: L1, L2, T1, C1a, T2	new theory
IV	Range & feasibility: P1 (both parts)	new theory
V	Cost model & profitability: T3, T4, T5	new theory
VI	Architecture: asymmetric prox unit, weight-stationary array, saturating datapath, the cast overhead as a first-class cost	RTL

§	Content	Source
VII	Validation: bit-exact model vs RTL; wordlength sweep vs T2; four-build synthesis vs T4; BER vs SNR; throughput/area vs TASER	measurements
VIII	Discussion: the negative result, and what it implies for ASIC vs DSP-FPGA	—

Note §V before §VI: the cost model motivates the architecture, rather than the architecture being presented and then justified after the fact.

5. Claims and defenses

Claim 1 — Proximal operator precision is governed by the *degenerate set*, not by contraction

Defense. Theorem 1, validated by direct measurement of the error decomposition. **Anticipated attack.** "This is just standard quantization noise analysis." **Response.** The novelty is the annihilation term: on D the operator destroys the perturbation exactly, so effective noise power scales with $|D^c|/n$. Standard analyses assume every element contributes $q^2/12$. Show the measured gap.

Claim 2 — No fixed wordlength ordering exists; we give the condition

Defense. Theorem 2 plus the four-way sweep showing the crossover. **Anticipated attack.** "Then the reconfigurable hardware can't be sized." **Response.** It can — sized for the *worst case over the operating envelope*, which Theorem 2 makes computable in closed form instead of by search. That is the practical contribution.

Claim 3 — Asymmetric pruning pays only when arithmetic is in fabric

Defense. Theorem 4 plus the four-build table: +37.1% / -2.9% from one formula. **Anticipated attack.** "This is a Vivado inference artifact, not a result." **Response.** Correct, and that is the point — it is a *property of the target*, which is exactly what a co-design methodology must account for. We control for it explicitly with `USE_DSP_L2` rather than letting the tool decide silently. This is the strongest defense in the paper: we found the confound and isolated it.

Claim 4 — Bit-exact model / RTL equivalence

Defense. Both builds pass element-wise against the golden model for all three operators. Cheap to state, and it is what makes every other number credible.

Claim 5 — Competitive area/throughput

Defense. Deferred. Needs F_{max} sweep and Mb/s normalisation before it can be claimed at all. **Do not write this section until §7 items 3-4 are done.**

6. Threats to the paper

Threat	Severity	Mitigation
Reviewer: "WLO is a solved, mature field"	high	Position against <i>search-based</i> WLO; our contribution is closed-form + the profitability condition, which search cannot express
Theorem 5 fails to compose	medium	Demote to discussion; T2 and T4 stand alone
Small array (N=8) reads as toy	medium	N is users , not antennas — 8 users × 32+ BS antennas is a real massive-MIMO config. Say it in §II, not in rebuttal
Real-valued only	medium	State the real-valued equivalent model explicitly; note complex doubles N
Fmax ~59 MHz vs TASER 232 MHz	high	Sweep the constraint; report throughput/area, not raw Fmax. If it stays low, own it as an edge/low-power operating point
Power numbers are vectorless	low	Regenerate from SAIF

7. Immediate next actions

Instrument the Theorem 1 error decomposition in `fxp_admm.py`. **Blocks all theory work.**
Add `d_Box / d_L1` measurement to `sweep_ordering.py`; locate the crossover.
Re-derive lane widths from Theorem 2; re-run the four-build synthesis on the justified widths.
Fmax sweep (tighten the constraint until failure) → real throughput.
SAIF-based power.
BER vs SNR, final widths.
Only then: write.

APPENDIX A — Theorem 1 validation (RUN, PASSED)

`model/error_decomp.py` and `model/annihilation.py`.

A.1 Three-term decomposition at the operating point

38,400 samples of $v = x + u$ harvested from float ADMM, lane swept $F = 4..14$. Reference is P_{float} applied to the same Q2.16 input with exact parameters, so parameter quantization registers as lane error.

operator	d = D /n T1 pred / measured standard model / measured		
L1 soft-threshold	0.000	0.99	0.61
Box projection	0.183	1.00	0.95
L2 ridge scaling	0.000	0.88	0.21

The standard flat- $q^2/12$ model is wrong by 4.8x for L2. Theorem 1 is within 12% there and within 1% for L1 and Box.

For L2 the discrepancy is entirely the coefficient term: contraction attenuates the *input* noise by gamma, but gamma's own quantization error is multiplied by v and is not attenuated at all. That is the term the original manuscript's "strict contraction implies minimum wordlength" argument omitted, and it is why L2 never measured as the cheapest lane.

A.2 Annihilation — the central mechanism (decisive test)

Parameter swept so d ranges over [0, 1], with parameters snapped to the lane lattice to zero the theta term and isolate annihilation. F = 8, so the standard model predicts a flat 1.2716e-06 at every d.

d	L1 measured MSE (1-d)q²/12		d	Box measured MSE (1-d)q²/12	
0.000	1.283e-06	1.272e-06	0.000	1.283e-06	1.272e-06
0.172	1.069e-06	1.053e-06	0.120	1.132e-06	1.119e-06
0.485	6.687e-07	6.543e-07	0.515	6.146e-07	6.172e-07
0.880	1.510e-07	1.525e-07	0.828	2.145e-07	2.188e-07
1.000	0.000	0.000	0.999	1.408e-09	1.351e-09

Over d in [0.12, 0.88]:

	T1 / measured standard / measured	
L1	0.991	3.838
Box	1.004	3.040

At d = 1 the measured error is **exactly zero**. The operator annihilates the perturbation completely. The standard model still predicts 1.27e-06 there, and is wrong by up to 900x at d = 0.999.

Theorem 1 is validated. Write it.

A.3 Two findings that came out of the validation

Design rule — parameters must lie on the lane lattice. A Box bound or L1 threshold that is not a multiple of the lane's quantization step contributes a constant error on every degenerate element, exactly cancelling the benefit of degeneracy. In the first run, alpha-scaling put the bound off-lattice and the Box annihilation benefit vanished entirely. This is a free, actionable result: constrain alpha to a power of two.

Revised explanation of why L1 is expensive. Not active-set misclassification (measured at zero, M6). At the operating point $d_{L1} = 0.000$ — the threshold $\kappa = 0.1$ is far below the signal level, so *nothing* is zeroed and there is no annihilation at all. The remaining gap to Box is the κ quantization term on D^c . This predicts that placing κ on the lattice should make L1 as cheap as Box at equal d . **Test this next — it is a strong, cheap, falsifiable prediction and it would be the first design rule the theory produces.**

A.4 Consequence for Theorem 2

The ordering is now fully derived rather than asserted:

$$W^*_{op} = \frac{1}{2} \log_2 [((1-d_{op})(L_{op}^2 q^2/12 + \text{theta terms on } D^c) + d_{op} (\text{theta terms on } D)) / \text{tolerance}]$$

The measured ordering (Box cheapest, L1 and L2 comparable) follows directly from $d_{Box} = 0.183 > d_{L1} = 0.000$ together with L2's un-attenuated coefficient term. No fixed inequality among the three survives, exactly as Theorem 2 states.